

UNIVERSIDAD REGIONAL AUTONOMA DE LOS ANDES



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Evaluación 1

$$y = \sqrt{(3x^2 - x)^3}$$

$$y = \sqrt{(3x^2 - x)^3} \quad \frac{d}{dx} (u^n) = n u^{n-1} \cdot \frac{du}{dx}$$

$$y' = \frac{1}{2} ((3x^2 - x)^3)^{-1/2} \cdot \frac{d}{dx} ((3x^2 - x)^3)$$

$$\frac{d}{dx} ((3x^2 - x)^3) = 3 (3x^2 - x)^2 \cdot \frac{d}{dx} (3x^2 - x)$$

$$\frac{d}{dx} (3x^2 - x) = 6x - 1$$

$$\frac{d}{dx} ((3x^2 - x)^3) = 3 (3x^2 - x)^2 \cdot (6x - 1)$$

$$y' = \frac{1}{2} ((3x^2 - x)^3)^{-1/2} \cdot 3 (3x^2 - x)^2 \cdot (6x - 1)$$

$$y' = \frac{3 (3x^2 - x)^2 (6x - 1)}{2 \sqrt{(3x^2 - x)^3}}$$

- Encuentre la derivada de $h(x) = (2x + 1)^5 (3x - 2)^7$.

$$h(x) = (2x + 1)^5 (3x - 2)^7$$

$$u' = 5(2x + 1)^4 \cdot 2 = 10(2x + 1)^4$$

$$v' = 7(3x - 2)^6 \cdot 3 = 21(3x - 2)^6$$

$$h'(x) = 10(2x + 1)^4 (3x - 2)^7 + 21(2x + 1)^5 (3x - 2)^6$$

Encuentre la derivada de $h(x) = \sec(4x^5 + 2x)$

$$h(x) = \sec(4x^5 + 2x)$$

$$\frac{du}{dx} = 20x^4 + 2$$

$$h'(x) = \sec(4x^5 + 2x) \tan(4x^5 + 2x) \cdot (20x^4 + 2)$$

$$h'(x) = \sec(4x^5 + 2x) \tan(4x^5 + 2x) (20x^4 + 2)$$

Calcular la derivada de: $f(x) = (7x^3 - 5)^4$

$$f(x) = (7x^3 - 5)^4$$

$$u = 7x^3 - 5$$

$$\frac{du}{dx} = 21x^2$$

$$f'(x) = 4(7x^3 - 5)^3 \cdot 21x^2$$

$$f'(x) = 84x^2(7x^3 - 5)^3$$

Calcular la derivada de: $g(x) = 3 / (4x^2 - 1)^4$

$$g(x) = 3 / (4x^2 - 1)^4$$

$$g(x) = 3 (4x^2 - 1)^{-4}$$

$$g'(x) = 3 \cdot (-4) (4x^2 - 1)^{-5} \cdot \frac{d}{dx} (4x^2 - 1)$$

$$\frac{d}{dx} (4x^2 - 1) = 8x$$

$$g'(x) = 3 \cdot (-4) (4x^2 - 1)^{-5} \cdot 8x$$

$$g'(x) = -96x (4x^2 - 1)^{-5}$$

$$g'(x) = -96x / (4x^2 - 1)^5$$

Aplicando la regla de los 4 pasos

$$F(x) = x^3 - 2x$$

$$F'(x+h) - F(x) = (x+h)^3 - 2(x+h) - (x^3 - 2x)$$

$$F'(x+h) - F(x) = (x^3 + 3x^2h + 3xh^2 + h^3) - 2(x+h) - (x^3 - 2x)$$

$$F'(x+h) - F(x) = x^3 + 3x^2h + 3xh^2 + h^3 - 2x - 2h - x^3 + 2x$$

$$F'(x) = 3x^2h + 3xh^2 + h^3 - 2h$$

$$F'(x) = \frac{3x^2\cancel{h} + 3x\cancel{h}^2 + \cancel{h}^3 - 2\cancel{h}}{\cancel{h}}$$

$$F'(x) = 3x^2 + 3xh + h - 2$$

$$F'(x) = 3x^2 + 3x\cancel{h} + \cancel{h} - 2$$

$h \rightarrow 0$

$$F'(x) = 3x^2 - 2$$